

# Term Projects: AI Programming on GPU and Microcontroller

## Project Overview

This term project is designed to provide practical experience in hardware programming for artificial intelligence (AI) applications. You will leverage the hardware architecture to program and optimize AI models on graphical processing units (GPUs) and microcontrollers (MCUs). The project is divided into two parts: the first focuses on training a Lenet Convolutional Neural Network (CNN) on the Jetson Nano using Cuda, and the second focuses on deploying the Lenet model on the MSP430FR5994 microcontroller. The MCU platform is MSP430FR5994, and you will need to program using Code Composer Studio (CCS).

### Part 1: Training Lenet on Jetson Nano with Cuda (60 + 10)

- 1) **Model Setup and Training:** The Lenet CNN model structure and dataset will be provided for this project. Please implement and train the model on Jetson Nano using Cuda. Your implementation should be able to complete training and produce the final testing accurately.
- 2) **Training Performance Evaluation:** Please measure and record the training time and the final testing accuracy. Then, observe and discuss the results with the team. If the results do not meet expectations, please identify potential problems and come up with improvement methods to optimize performance further.

**Bonus for Part 1 (+10):** Additional bonus will be given for applying GPU parallelism and further optimizing the training process through improved kernel design and implementation. The degree of parallelism and optimization will impact the bonus score.

### Part 2: Deployment of Lenet on MSP430FR5994 (40 + 10)

- 1) **Sequential Inference and Model Size Analysis:** Please first complete the sequential inference logic of the Lenet model on MSP430FR5994. After that, observe the model size and memory usage to determine whether quantization is needed for deployment on the MCU.
- 2) **UART-Based Image Input and Result Output:** Please design the UART communication code so that an input image with 8-bit pixel values can be sent from the computer to MSP430FR5994. The MCU should receive the image, perform the inference, and return the classification result to the computer interface through *uartassist* (on canvas).
- 3) **Deployment Performance Evaluation:** Please measure and record the average execution time and inference accuracy of the deployed model on MSP430FR5994. Then, observe and discuss the results with the team. If the results do not meet expectations, please identify potential problems and come up with improvement methods to optimize performance further.

**Bonus for Part 2 (+10):** If quantization is needed, quantization-aware training is recommended over post-training quantization to better preserve accuracy. Additional bonus will also be given to utilize the MCU's onboard accelerators, such as LEA and DMA, to further improve inference performance.

## Project Report

In the end, each group is required to submit a comprehensive report detailing the experimenting steps, code optimizations made, and an analysis of the performance with figures and discussions. The report should also include discussions on any challenges faced and how they were addressed. Additionally, all source code (GPU and MCU) must be submitted along with the report.